Tales Boratto

Data Scientist

E-mail: tboratto@gmail.com | Phone: +55 (32) 999726978 | LinkedIn | GitHub | Google Scholar

Summary

Data Scientist with one year of industry experience developing, evaluating, and deploying machine learning solutions for real-world problems. PhD candidate in Computational Modeling with a strong background in Python, machine learning, statistics, and predictive modeling. Combines industry experience with academic research to prepare data, engineer features, optimize models, and conduct rigorous experimental evaluations.

Work Experience

Mid Data Scientist II

TRACTIAN | Brazil | October 2025 - October 2026

Developed an automated harmonic series detection algorithm using signal processing and feature engineering, improving detection accuracy by 20% while maintaining false positive rate at 1%.

Engineered domain-informed features for a mechanical looseness detection model, reducing the false positive rate by approximately 5% and improving the reliability of automated alerts.

Deployed machine learning models to production through existing GitHub Actions workflows and monitored their post-deployment performance.

Identified a model issue affecting approximately 4% of the dataset through error analysis and proposed corrective measures.

Developed interactive Streamlit dashboards to inspect predictions and investigate individual cases, supporting model behavior analysis and error investigation.

Designed and implemented a database and key performance indicators (KPIs) to track production model performance, supporting data-driven model updates and improvements.

Prepared, cleaned, and annotated datasets, applying signal processing and feature engineering, and ran hyperparameter optimization tracked with MLflow to support reproducible model development and evaluation.

Research Experience

Doctoral Researcher in Computational Modeling

Federal University of Juiz de Fora | Brazil | December 2022 - Present

Develop and evaluate federated learning approaches in Python for anomaly detection in time-series data, investigating model performance across distributed learning settings.

Design comparative machine learning experiments using scikit-learn, XGBoost, and autoencoder to assess how feature representations and learning configurations affect predictive performance.

Contributed to five journal articles and two international conference papers during doctoral research, communicating methodological developments and experimental findings to scientific audiences.

Investigated federated learning for healthcare applications during a research exchange at the University of Nottingham (UK), collaborating with a multidisciplinary team.

Graduate Researcher – Master's in Computational Modeling

Federal University of Juiz de Fora | Brazil | March 2020 - November 2022

Developed an experimental audio data collection workflow and evaluated 60 classification configurations across five feature sets, achieving 99.75% mean accuracy with logistic regression using 24 selected features on a controlled experimental dataset.

Reduced feature dimensionality from 30 to 5 through feature selection, an 83.3% reduction, with performance metrics within one percentage point of the higher-performing 24-feature configuration. Applied statistical testing and error analysis to inform model selection.

Education

PhD in Computational Modeling – In Progress
Federal University of Juiz de Fora | Brazil | Expected December 2026

Master's Degree in Computational Modeling
Federal University of Juiz de Fora | Brazil | November 2022

Bachelor's Degree in Mechanical Engineering
Federal University of Juiz de Fora | Brazil | January 2020

Bachelor's Degree in Exact Sciences
Federal University of Juiz de Fora | Brazil | July 2019

Technical Skills

Programming & Libraries: Python • pandas • NumPy • SciPy • scikit-learn • XGBoost • Matplotlib • TensorFlow • Seaborn • SQL

ML and Statistics: Machine Learning • Predictive Modeling • Classification • Regression • Clustering • Anomaly Detection • Time Series Analysis • Feature Engineering • Feature Selection • Hyperparameter Optimization • Hypothesis Testing • Statistical Analysis • Federated Learning

Data & Signal: Signal Processing • ETL • Data Cleaning • Data Processing • Data Annotation • Exploratory Data Analysis

MLOps & Tools: GitHub • Streamlit • Jupyter Notebook • MLFlow (experiment tracking)

Languages

Portuguese: Native.
English: Advanced.
Spanish: Intermediate.